

Zhalok Rahman

Software Engineer 2 at Datatree Technology

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LinkedIn: <https://www.linkedin.com/in/zhalok-rahman/>

Github: <https://github.com/zhalok>

Video Introduction: <https://www.loom.com/share/c5ab46a4737244edaf9f052555767a1>

Skills

Golang, Python (FastAPI, Asyncio, Pytorch, Pandas), Node JS (Nest JS, Express JS, Koa), JavaScript/TypeScript (React JS, Next JS, PostgreSQL, MongoDB, Redis (Cache, Pub/Sub, Streams), RabbitMQ, WebSocket, SSE, Docker, AWS (ECS, CloudWatch, SQS, Kinesis, Lambda, OpenSearch, DynamoDB), Grafana, Prometheus, CI/CD (Github Actions, CodePipeline)

Education

Shahjalal University of Science and Technology

Bachelors of Science in Computer Science

2019-01-01 – 2024-04-01

Experience

Kinetik Healthcare

Software Engineer 2

2025-04-01 – Present

- Built **Florence** an **AI voice assistant** which handles around 80 percent of our ride initiation requests
- Developed internal **VoIP** testing tools for simulating calls for the voice agent
- Built a source control pipeline to manage **11Labs agent as code**
- Managed **CI/CD pipeline** for deploying agent
- Integrated Intercom based human agent call center with AI call center for transferring calls as fallback
- Built **Grafana dashboard** for monitoring the AI agent performance
- Contributed to **Langgraph** based orchestrator service for a large scale **AI Agent**
- Enhanced realtime notification pipeline with push based mechanism with **Redis Pub/Sub, CDC, Redis Streams**
- Migrated legacy code from **Express JS** to **Nest JS** for cleaner domain driven design
- Enhanced existing **RBAC** system with realtime access revocation
- Investigated production incidents based on logs and traces from **CloudWatch** and **Grafana**
- Optimized MongoDB query performance by **41 percent** using covered atlas search index scan
- Reduced network content download time by **66 percent** with default query projection
- Conducted AI-driven development using **Kiro**

PropertyFinder

Software Engineer

2024-10-01 – 2025-10-31

- Ensured reliability and performance of **Golang** based B2B platform
- **Traced race conditions** between distributed services that was causing p1 incidents and confusing user experiences
- Investigated and fixed root cause of critical production incidents using metrics, logs, and traces from **Grafana**
- Conducted **Load Testing** and built dashboards to observe the performance of the entire system under high stress
- Written scripts for data fix to mitigate incidents
- Contributed to the modernization of B2B service, making architectural decisions and strategies for efficient migration
- Written unit tests to ensure **80 percent coverage**
- Designed asynchronous event-driven pipelines using **SQS** and **Kinesis Streams**
- Configured alerts in **Grafana** with **Slack** integration for real-time notification on any potential incident
- Used **Windsurf** for studying legacy codebases efficiently

Learnyx AI

Fullstack Engineer

2023-10-01 – 2025-02-01

- Architected and built a scalable REST API with **NestJS**
- Designed an AI generation pipeline using **LangChain** for structured prompt management and multi-step chaining

- Integrated **Langfuse** for LLM observability, enabling cost tracking and performance tracing across model calls
- Decoupled heavy workloads from the main request cycle using asynchronous **RabbitMQ** consumers, improving resource efficiency
- Enhanced UX by implementing real-time streaming of AI-generated responses
- Optimized load performance using a hybrid lazy-loading and preloading strategy
- Acted as a technical advisor in stakeholder discussions, bridging engineering constraints with business goals
- Mentored a team of 3 engineers on system architecture patterns and backend best practices
- Managed self-hosted infrastructure on **AWS EC2** using **Coolify**, supporting **500+ active users**
- Integrated **Stripe** to handle recurring billing, subscription lifecycle, and payment management

Projects

MongoDB vs PostgreSQL benchmarking for storing documents

Benchmark harness comparing MongoDB BSON and PostgreSQL JSONB for document storage under concurrent load, with container and database-internal metrics.

<https://github.com/zhalok/mongodb-bson-vs-postgres-jsonb-benchmarking>

- Benchmark results: <https://bit.ly/4dbwS4h>

Tech: Golang, JavaScript, Docker, PostgreSQL, MongoDB, Prometheus, Grafana, Loki, Promtail

PostgreSPECTOR

Observability toolkit for PostgreSQL that exposes execution stats and container metrics to Prometheus and Grafana, with logs shipped via Loki.

<https://github.com/zhalok/PostgreSPECTOR.git>

- Demo: <https://www.youtube.com/watch?v=w2OCn0X3bRw>

Tech: Golang, Prometheus, Grafana, PostgreSQL, Loki, Promtail, Docker

Obstainer

Lightweight container observability tool that reads metrics straight from cgroups and exposes them to Prometheus and Grafana.

<https://github.com/zhalok/obstainer>

- Demo: <https://lnkd.in/p/gw6QYyMr>

Tech: Golang, Grafana, Prometheus, Docker, Loki, Promtail

Data Stores Python

Experiments comparing TCP server I/O models (thread-per-connection, non-blocking, event-driven) under load.

<https://github.com/zhalok/data-stores-python>

- Benchmark results: <https://bit.ly/3V6UUr9>

Tech: Python, Socket, System Calls, Docker

Go Http From Scratch System Calls

HTTP server implemented from raw sockets and system calls in Go, without the net/http package.

<https://github.com/zhalok/go-http-from-scratch-system-calls> Tech: Golang, Syscalls, OSSignals, Sockets

Certifications

AWS Certified Cloud Practitioner

AWS

<https://www.credly.com/badges/201260d3-12e0-4d22-a49f-4e64d68b348e>

Deep dive on AWS services

Optimizing MongoDB Performance with Tuning Tools

MongoDB

<https://www.credly.com/badges/e861b7b4-028a-47d4-aed6-7e586d4d95da>

Learned about diagnosing and tuning MongoDB performance using database tooling

Fundamentals of Operating Systems

Udemy

<https://www.udemy.com/certificate/UC-ff0d44a3-8917-40cd-aef9-a01e258aee49/>

Learned about Process structure, Memory management, CPU cycles, I/O management

Fundamentals of Network Engineering

Udemy

<https://www.udemy.com/certificate/UC-0a49d23e-036a-4c80-ad5a-410ccb877ebb/>

Learned about under the hood mechanisms of network protocols

Fundamentals of Database Engineering

Udemy

<https://www.udemy.com/certificate/UC-afe8cbda-1420-4cca-a833-aed013e7a6e2/>

Learned about ACID, Data organization, Indexing, Partitioning, Sharding, Concurrency control, Replication, Sharding, Partitioning etc